Fabrication and Integration Process

1. Material Choice Justification

Foam Board (Primary Structure with Interlocking System)

Foam board was used as the primary material along with an **interlocking joint system**.

Why this was chosen:

- **Lightweight and easy to cut**, allowing precise slot creation
- Enabled a **modular construction approach**
- Interlocking reduced dependency on adhesives
- Allowed **easy assembly, disassembly, and correction**

Compared to MDF, the foam board provided faster iteration while still supporting **structural stability through interlocking geometry**.

Sticker Paper (Surface Detailing)

- Used to create a **cohesive sci-fi / quarantine lab aesthetic**
- Eliminated the need for paint
- Allowed **clean, high-resolution textures** to be applied quickly

MDF (Laser-Cut Puzzle Components)

MDF was used specifically for **interactive and precise elements**:

- Allowed **accurate laser-cut detailing**
- Created contrast between structural and functional parts

Tape (Paper + Electrical)

- **Paper tape**: temporary alignment during assembly
- **Electrical tape**: wire management and visual cleanup

2. Fabrication Steps (Build Log)

Step 1: Concept Layout & Planning

- Defined spatial zones: door, puzzle panel, desk, interaction areas
- Planned placement of electronics and hidden mechanisms

Step 2: Designing Interlocking System

- Created slot-and-tab joints for foam board panels
- Ensured tight fits for:
    - Walls
    - Base
    - Structural supports

This step replaced the need for heavy adhesives and increased precision

Step 3: Cutting and Assembly

- Foam board pieces were cut manually
- Interlocking edges were fitted together to form:
    - Walls
    - Floor
    - Furniture elements
- Structure became **self-supporting due to joint system**

Step 4: Surface Application

- Sticker paper applied after structural assembly
- Helped hide joints and enhance realism

Step 5: Laser Cutting MDF Components

- Designed and cut puzzle interfaces and clue elements
- Integrated into foam board structure at key interaction points

Step 6: Electronics Integration

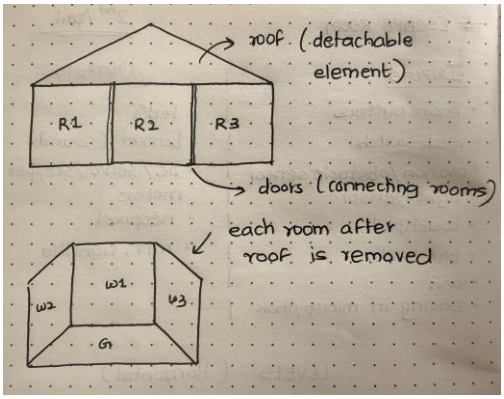
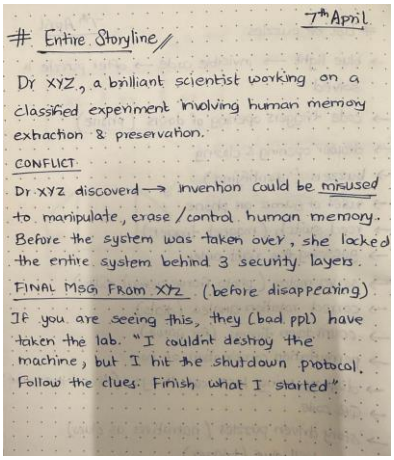
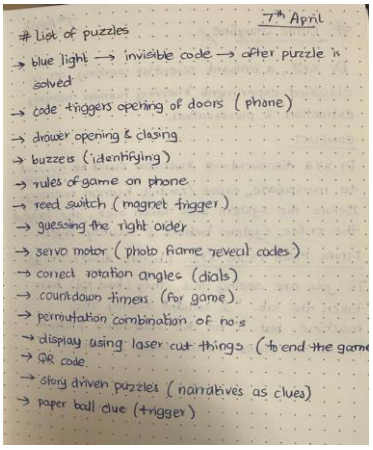
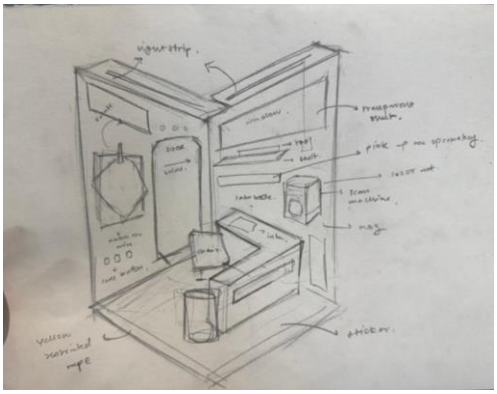
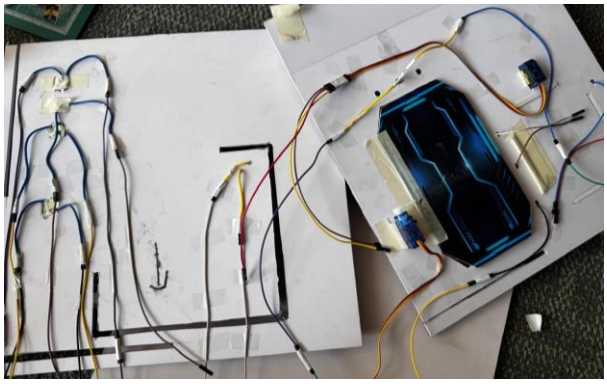
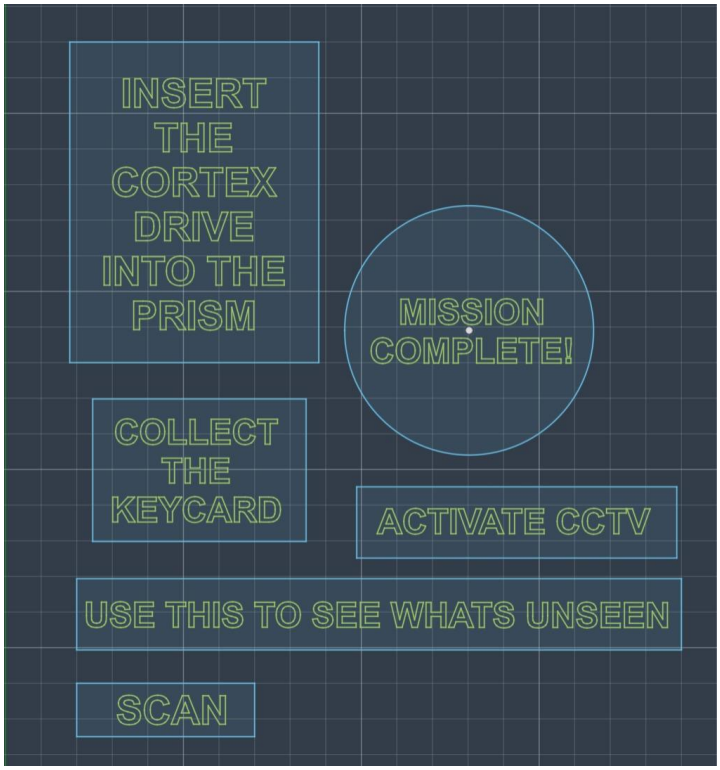
- Installed:
    - Servo behind door
    - IR sensor near interaction zone
    - Push buttons on floor panel
- Wiring routed along edges and concealed using tape

Step 7: Mechanism Testing & Iteration

- Tested servo movement with door
- Adjusted:
    - Angle calibration
    - Sensor sensitivity
    - Trigger logic

Step 8: Final Assembly & Detailing

- Placed narrative elements (notes, warning signs, clues)

- Ensured visual coherence with theme
- Final checks for interaction flow

A few glimpses of our fabrication process: -